Michael C.

Java Backend Engineer · Payments & Wallets · Distributed Systems

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Open to remote roles

SUMMARY

Hands-on Java backend engineer with 13 years of experience, including Alibaba Group and DiDi, followed by wallet and payment systems at Huobi and dtcpay.
For the past 4 years I've owned backend modules for wallets and payments: a non-custodial multi-chain wallet at Huobi and a consumer e-wallet at dtcpay in Singapore. My focus is transaction integrity, reliability, and recovery from failures.
I take a module from design to production on my own and work well in async, remote teams across time zones.

HOW I WORK

Written-first: requirements and decisions live in docs; async by default, meetings short and purposeful.
Pragmatic: judged by working results and data; focused on solving problems.
Professional process: code review, clear standards, clear ownership.
Evidence-led: use measurements, production observations, and technical trade-offs to resolve disagreements.
Transparent collaboration: document decisions, share context, and raise issues directly and respectfully.

SKILLS

Java / Spring Boot: thread pools, JVM OOM troubleshooting, transaction pitfalls
MySQL: index design, cursor pagination, conditional updates + row locks
Redis: caching, distributed locks, Bloom filters
Kafka: reliable delivery, idempotent consumers
Payments: idempotency, state machines, channel integration, reconciliation
Blockchain: wallet creation, transaction broadcast, block scanning, reorg handling

EXPERIENCE

dtcpay — Senior Java Backend Engineer — E-Wallet Owner

Singapore-based digital payments company | Apr 2024 – Present | Singapore
Website: dtcpay.com

MAS-licensed payment institution (fiat + stablecoin). Owned the core flow of the consumer e-wallet app: top-up → withdrawal → daily reconciliation; every balance change is booked by the wallet service.

- Concurrency safety (no overdraft): a single conditional UPDATE checks and deducts the balance under a row lock, so concurrent debits queue up and never go negative.
- Idempotent APIs: one-time Redis tokens block duplicate submits; a unique index on the business order ID is the final guard.
- Lost-callback recovery: verify signature and amount on channel callbacks; scheduled jobs query the channel to recover missed notifications.
- Idempotent payouts: freeze funds before payout; on channel timeout only query, never resend; deduct on success, unfreeze on failure.
- Eventual consistency: debit and pending notification are written in one DB transaction, then delivered via Kafka with retries (local message table).

Huobi — Senior Java Backend Engineer — Blockchain Wallet Owner

Global crypto brand — Huobi Wallet team | May 2022 – Apr 2024
Website: www.htx.com

Huobi Wallet: a non-custodial multi-chain wallet with ~12M users (private keys stay on the user's phone). Owned the core flow: wallet creation → on-chain transfer → deposit detection → deposit notification.

- Chain reorgs: detect broken parentHash links, roll back and rescan; credit deposits only after enough confirmations.
- OOM fix: the block scanner crashed while catching up; root-caused an unbounded queue holding blocks in memory and switched to a bounded queue with back-pressure.
- Reliable delivery + idempotent consumption with Kafka: persist notifications before sending, retry failures, and deduplicate repeated deposit events.
- Address filtering with a Bloom filter: discard non-user addresses in each block before hitting the database.

• Scanner HA with a Redis distributed lock: one active scanner, automatic failover, resume from the saved cursor.

DiDi — Senior Java Backend Engineer

Major global ride-hailing platform — China's Uber | Oct 2018 – May 2022 | Hangzhou, China
Website: www.didiglobal.com
Ride-hailing platform. Tech lead of the company-wide search service.

• Built the search service from 0 to 1 on Elasticsearch.
• Built SQL lineage parsing that automatically extracts upstream/downstream table dependencies, exposed as a query service.
• Provided unified query APIs consumed by systems across the company.

Alibaba Group — Java Backend Engineer — Distributed Job Scheduling

Global e-commerce and cloud technology group | Oct 2016 – Oct 2018 | Hangzhou, China
Website: www.alibabagroup.com
Alibaba Group. Core developer of the scheduling system behind Alibaba Cloud DataWorks: master election, job sharding and timed triggers.

• Master HA: nodes compete for a database row lock (SELECT … FOR UPDATE); when the master dies the lock is released and another node takes over automatically.
• Job sharding: machines map to a fixed set of logical nodes, so scaling in/out only changes the mapping (similar to consistent hashing).
• Quartz-based scheduling of 1M+ hourly/daily/weekly/monthly jobs; average job development time dropped from 36h to 17.45h after launch.

Ourpalm — Java Backend Engineer

Publicly listed mobile game developer and publisher | Mar 2013 – Oct 2016 | Beijing, China
Website: www.ourpalm.com
Mobile game company listed on the Shenzhen ChiNext board. Built the shared server side used by multiple games: log collection SDK and operations admin.

• Designed and built a server-side game log collection SDK adopted by multiple mobile games.
• Developed and maintained the Java operations admin system and its scheduled jobs.

PERSONAL PROJECTS (AI AGENT)

Homepage: Personal homepage built and launched solo with AI agents (Next.js + Vercel) · michael.opc-x.org
Quarry: Multi-channel data collection: scrapes job postings from BOSS, RemoteOK, V2EX, Telegram, X and more into one feed, with automatic tagging and scoring · quarry.opc-x.org

EDUCATION

Hawaii Pacific University — M.S. Artificial Intelligence · GPA 3.88 / 4.0 · With Distinction — Jan 2025 – Jul 2026

University of Electronic Science and Technology of China — B.Eng. Computer Science and Technology — Sep 2008 – Jul 2012